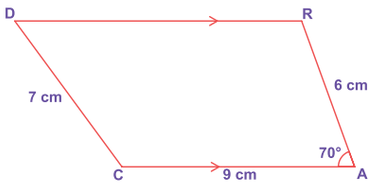
5.11.2 Constructing a trapezium when its three sides and one angle are given

Example 5.27

Construct a trapezium **CARD** in which $\overline{CA}$ is parallel to $\overline{DR}$, CA=9*cm*, $\angle CAR = 70°$, AR=6*cm* and CD=7*cm*. Also find its area.



Rough diagram

Solution:

Given:

CA=9*cm*, $\angle CAR = 70°$ AR=6*cm*, and CD=7*cm* and $\overline{CA} \parallel \overline{DR}$

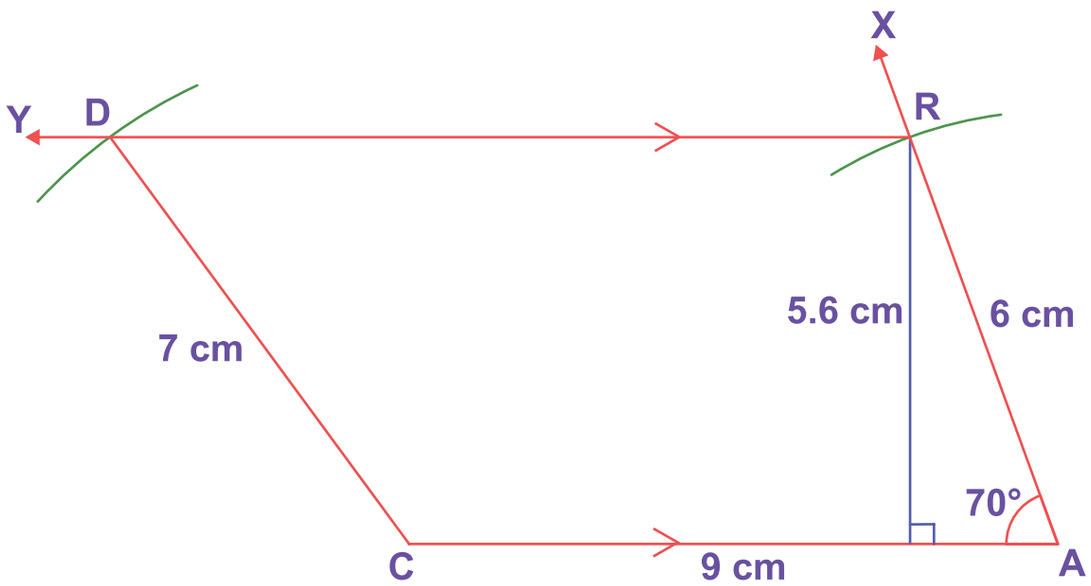


Fig. 5.46

Steps:

1. Draw a line segment CA= 9*cm*.
2. Construct an angle $\angle CAX = 70°$ at A.
3. With A as centre, draw an arc of radius 6*cm* cutting AX at R.
4. Draw RY parallel to CA.
5. With C as centre, draw an arc of radius 7*cm* cutting RY at D.
6. Join CD. CARD is the required trapezium.

Calculation of Area:

Area of the trapezium CARD $= \frac{1}{2} \times h \times (a+b)\, sq.units$

$= \frac{1}{2} \times 5.6 \times (9+11) = 56\, sq.cm$

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